

A4: Calculating necessary ARR values to create a 5k Hz square wave using 4 MHz input clock

Clock timer: $f_{TIM2} = 4 \text{ MHz}$

A single clock tick: $T_{\text{tick}} = \frac{1}{\text{input clock}} = \frac{1}{4,000,000} = 0.25 \mu\text{s}$

Desired output: $f_{\text{out}} = 5 \text{ kHz}$ Period: $\frac{1}{5000} = 200 \mu\text{s}$

$$\frac{\text{Period}}{\text{Single Clock tick}} = \frac{200 \mu\text{s}}{0.25 \mu\text{s}} = 800 \text{ ticks} \quad \star \text{ Count includes 0 } \star$$

$$\text{ARR} = 799$$

- Auto-Reload Register (ARR)

- Sets the value where the timer period ends.

- For up-counting:

- Counts (CNT) from 0 to ARR

- Then an update event occurs + the counter restarts

- Capture/Compare Register (CCR)

- TIM2 has several (CCR1, CCR2, ...)

- For A4, TIM2 is in Compare mode

- When $\text{TIM2} \rightarrow \text{CNT} = \text{TIM2} \rightarrow \text{CCR1}$

- the timer can set a compare flag and request an interrupt

- Timer frequency = 4 MHz ARR = 799
- Timer Tick = 0.25 μ s CCR1 = 199

- For a 25% duty cycle, the pin is high for:

$$0.25(200 \mu\text{s}) = 50 \mu\text{s}$$

and low for remaining period. 150 μ s

(a) 0.25 μ s/tick

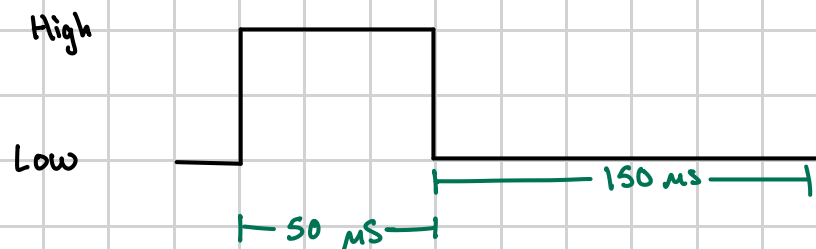
$$\frac{50 \mu\text{s}}{0.25 \mu\text{s}} = 200 \text{ ticks or counts}$$

$$\therefore \text{CCR1} = 199$$

- The timer counter frequency

$$f_{\text{CNT}} = \frac{f_{\text{TIM2}}}{\text{PSC} + 1}$$

*We want counter to stay @ full 4MHz



CNT: 0 199 799
 Period begins CCR1 Event occurs update Event